

Module 2

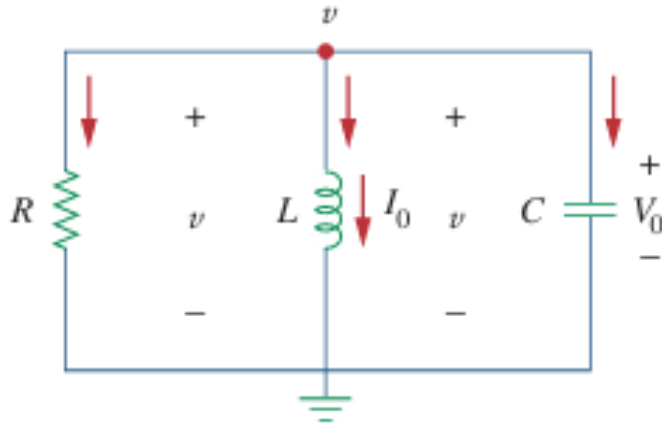
Transient Analysis of First and Second-Order Circuits

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Source Free RLC parallel circuit

Consider the parallel RLC circuit shown. Assume initial inductor current I_0 and initial capacitor voltage V_0 .



At $t = 0$,

$$i(0) = I_0 = \frac{1}{L} \int_{-\infty}^0 v(t) dt \quad \text{--- 1}$$

$$v(0) = V_0 \quad \text{--- 2}$$

Since the three elements are in parallel, they have the same voltage v across them. According to passive sign convention, the current is entering each element; that is, the current through each element is leaving the top node.

Applying KCL at the node,

$$\frac{v}{R} + \frac{1}{L} \int_{-\infty}^t v(\tau) d\tau + C \frac{dv}{dt} = 0 \quad \text{--- 3}$$

Taking the derivative with respect to t and dividing by C results,

$$\frac{d^2v}{dt^2} + \frac{1}{RC} \frac{dv}{dt} + \frac{1}{LC} v = 0 \quad \text{--- 4}$$

Source Free RLC parallel circuit

The characteristic equation is: $s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0$ --- 5

The roots of the characteristic equation are,

$$s_{1,2} = -\frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \quad \text{--- 6}$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \quad \text{--- 7}$$

Where,

$$\alpha = \frac{1}{2RC}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

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Overdamped Case ($\alpha > \omega_0$):

Considering Eqn.6 and $\alpha > \omega_0$: $L > 4R^2C$. The roots are negative and real. The response is,

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \text{--- 8}$$

Critically Damped Case ($\alpha = \omega_0$): Considering Eqn.6 and $\alpha = \omega_0$: $L = 4R^2C$. The roots are real and equal. The response is,

$$v(t) = (A_1 + A_2 t) e^{-\alpha t} \quad \text{--- 9}$$

Underdamped Case ($\alpha < \omega_0$): Considering Eqn.6 and $\alpha < \omega_0$: $L < 4R^2C$. The roots are complex.

$$s_{1,2} = -\alpha \pm j\omega_d$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

The response is,

$$v(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

Source Free RLC parallel circuit

The constants A_1 and A_2 and in each case can be determined from the initial conditions. $V(0)$ is found from eqn.2, $dv(0)/dt$ is found from eqn.3.

$$v(0) = V_0$$

$$\frac{V_0}{R} + I_0 + C \frac{dv(0)}{dt} = 0$$

$$\frac{dv(0)}{dt} = -\frac{(V_0 + RI_0)}{RC}$$

Having found the capacitor voltage for the parallel RLC circuit, we can readily obtain other circuit quantities.

Exercise

In the parallel circuit, find $v(t)$ for $t > 0$ assuming $v(0) = 5 \text{ V}$, $i(0) = 0$, $L = 1 \text{ H}$ and $C = 10 \text{ mF}$.

Consider $R = 1.923 \text{ } \Omega$, $R = 5 \text{ } \Omega$, $R = 6.25 \text{ } \Omega$.

$R = 1.923 \text{ } \Omega$

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 1.923 \times 10 \times 10^{-3}} = 26$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 10 \times 10^{-3}}} = 10$$

$\alpha > \omega_0$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -2, -50$$

$$v(t) = A_1 e^{-2t} + A_2 e^{-50t}$$

Now apply the initial conditions to get A_1 and A_2 .

$$v(0) = 5 = A_1 + A_2$$

$$\frac{dv(0)}{dt} = -\frac{v(0) + Ri(0)}{RC} = -\frac{5 + 0}{1.923 \times 10 \times 10^{-3}} = -260$$

$$\frac{dv}{dt} = -2A_1e^{-2t} - 50A_2e^{-50t}$$

At $t = 0$, $-260 = -2A_1 - 50A_2$

∴ $A_1 = -0.2083$ and $A_2 = 5.208$.

$$v(t) = -0.2083e^{-2t} + 5.208e^{-50t}$$

Next class: Step response of Series RLC circuit